

$\Rightarrow$ Show that the function in exercise 81 to 87 are all solution of the wave eq.

81

$$w = \sin(n+ct)$$

wave eq.

$$\left\{ \frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial n^2} \right\}$$

1, $\frac{\partial w}{\partial t}$ first derivative $\cos(n+ct) \cdot c$

2 $\frac{\partial^2 w}{\partial t^2}$ second order $-\sin(n+ct) c$

3 simplify $\frac{\partial^2 w}{\partial t^2} = -c^2 \sin(n+ct)$

Now prove,

$$\frac{\partial^2 w}{\partial t^2} = -c^2 \sin(n+ct) \quad (L.H.S.)$$

$$\frac{c^2 \partial^2 w}{\partial n^2} = c^2 (-\sin(n+ct)) \quad R.H.S.$$